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CYTOSKELETAL FILAMENTS AND CELLULAR JUNCTIONS

Cytoskeletal Filaments

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INTRODUCTION :

Cyto : Cell, Skeleton : Bone.

Support system : movement of cells (Gives strength).

Types :

Based on size (Diameter).

- Microtubules : Big.
- Microfilaments : Small.
- Intermediate filaments : Most abundant.



Microfilaments Microtubules Intermediate filaments

Cytoskeletal filaments

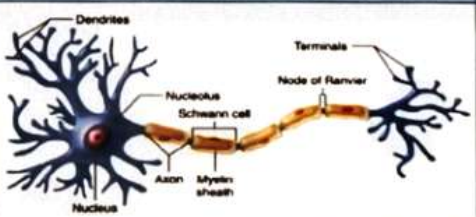
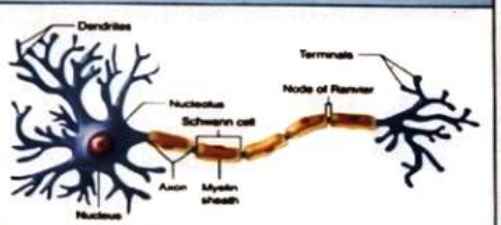
I. MICROTUBULES :

Proteins :

1. Kinesin.
2. Dynein.
3. Tubulin.

Functions :

1. Axonal transport (AT) :

	Forward AT	Reverse AT
Transport direction	 Cell body to synapse (Anterograde/ antegrade)	 Syapse to cell body
Protein	Kinesin	Dynein
Speed of transport	Fast : 400 mm/day Slow : 0.5–10 mm/day	Fast : 200 mm/day
Clinical significance	-	Invasion of microorganism : • Rabies virus. • Tetanus toxin : Tetanospasmin.

2. Ciliary motility :

- Physiological cilia motility :
  - In lung : Clearing sputum (Protective).
  - During embryological development : Rotation of internal organs.

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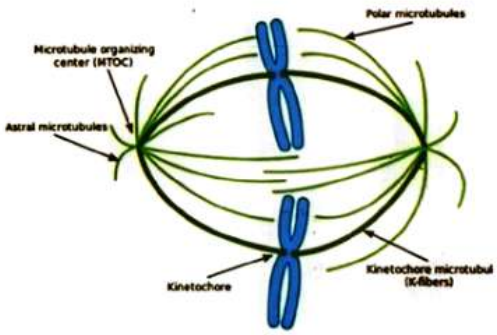
- Cilia structure :
  - 9+2 microtubule arrangement.
  - Dynein (Arm-like structure) : Cilia, sperm, fallopian tube motility.
- Clinical significance :

  Immotile cilia syndrome (Kartagener syndrome) : Absent dynein.
  - Bronchiectasis : Absent cilia motility → Sputum accumulated → Infections.
  - Infertility : Absent sperm & fallopian tube motility.
  - Situs inversus : Absent rotation of organs.

3. Chromosomal movement during cell division :
- Protein : Tubulin (In microtubule organising centre : MTOC).
- Significance : Movement of chromosomes into daughter cells during mitosis (For cell survival).
- Applied aspect : Microtubule inhibitor drugs.
  - Anti cancer drugs : Vinca alkaloids (Vincristine, Vinblastine), Taxanes (Paclitaxel).
  - Anti gout drugs : Colchicine.



Motile cilium



Microtubules during mitosis

a. MICROFILAMENTS :

Proteins :

1. Actin.
2. Myosin.

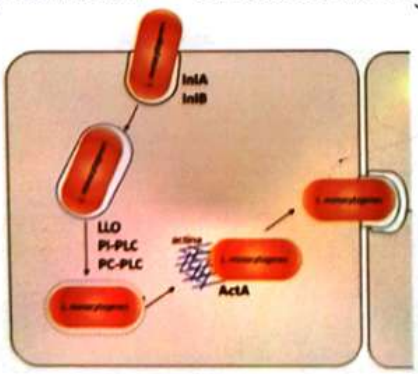


Skeletal muscle

Functions :

1. Skeletal muscle contraction & relaxation : Sliding filament theory.
2. Cell motility : Actin polymerization → Enhances motility.

Clinical significance :



Action of Listeria

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Listeria —Internalins (A/B)→ Enters into cell
↓
Engulfed into vacuole (To contain its spread)
Listeriolysin-O (LLO)
↓
Breakdown vacuole
Act-A
↓
Actin polymerization
↓
Comet tail pattern (Polymerized actin)
↓
Tumbling motility
↓
Infect nearby cells

3. INTERMEDIATE FILAMENTS :

Cell specific : used as tumor markers.

Protein	Location	Clinical significance
Keratin (Cytokeratin)	Epithelial cells	• Tumor marker in squamous cell carcinoma. • Mallory Denk bodies : Pathological deposition in liver in alcoholic liver disease.
Desmin	Muscle	Tumor markers in sarcoma (Rhabdomyosarcoma)
Vimentin	Connective tissues : Fibroblast	Markers for tumors of mesenchymal origin
Glial Fibrillary Acidic Protein (GFAP)	Astrocytes (Star shaped glial cells)	Tumor markers in glioma (Astrocytoma, Glioblastoma multiforme)
Lamin	Nucleus of cells	Progeria : Disorder of premature ageing (Werner syndrome)

Cellular Junctions

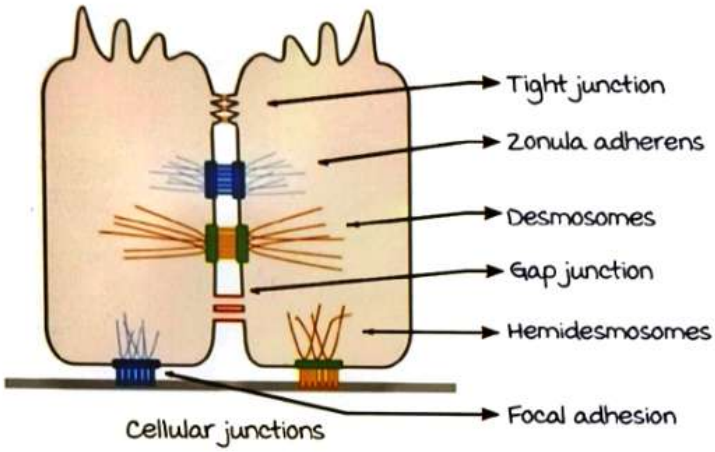
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Types
- Cell-cell junction
  - Tight junction
  - Cell adhesive junctions (Zonula adherens, Desmosomes)
  - Gap junctions
- Cell-basal lamina junction
  - Hemidesmosomes
  - Focal adhesions

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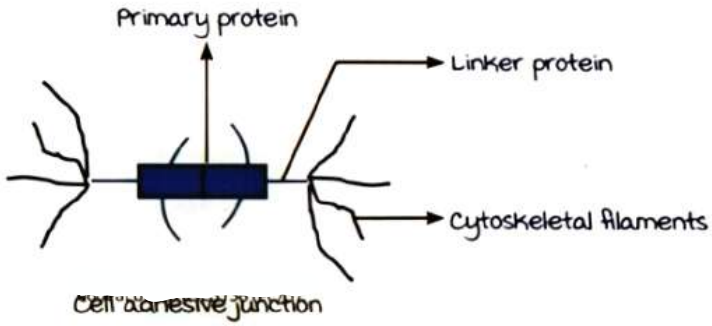
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CELL ADHESIVE JUNCTIONS :

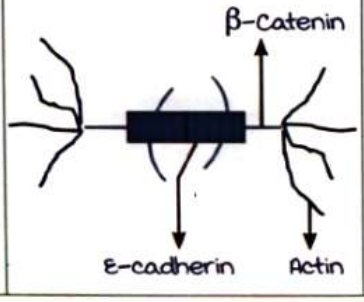
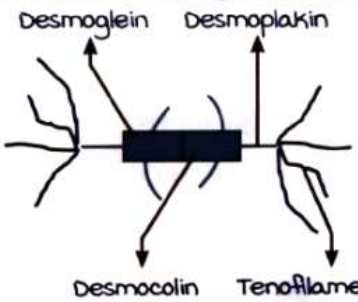
Uses cell adhesive molecules (CAMs) :

- Cadherins.
- Integrin.
- Selectin.



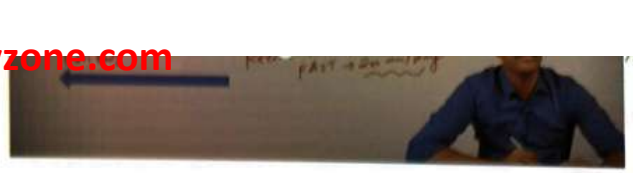
Structure :

- Primary protein.
- Linker protein.
- Cytoskeletal filaments.

	Zonula adherens	Desmosomes
Primary protein	E-cadherin	• Desmoglein. • Desmocolin.
Linker protein	β-Catenin	• Desmoplakin
Cytoskeletal filament	Actin	Tonofilaments (Intermediate filament)
Clinical significance	Molecular hijacking : Internalins (In Listeria) ↓ Recognizes E-cadherin ↓ Invade cells.	• Abundant in structures with mechanical stress (Heart, skin & uterine cervix). • Pemphigus disorders (Skin blistering disorder). • Autoantibodies against desmoglein.
		

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TIGHT JUNCTION :

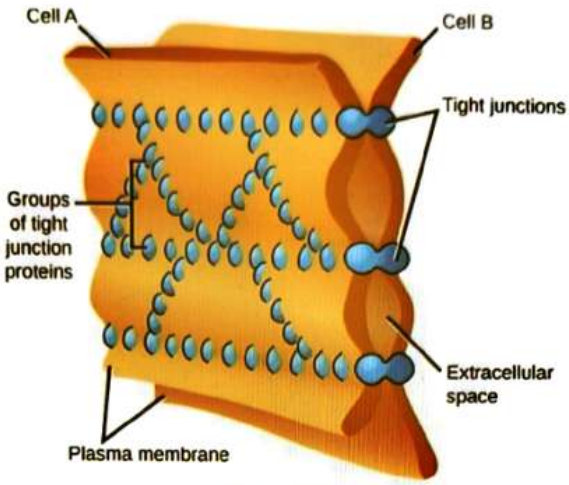
AKA Zona occludens.

Proteins :

1. Occludin.
2. Claudin.

Location :

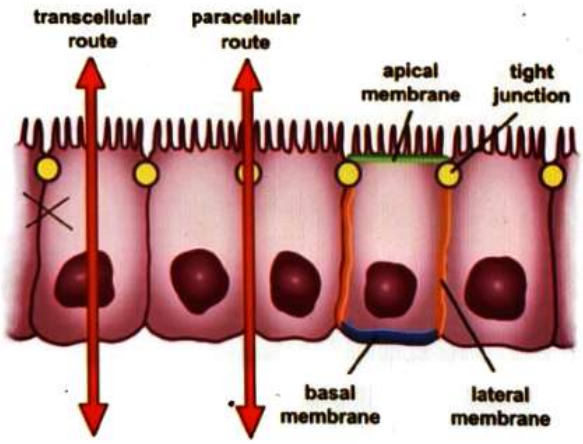
- Blood brain barrier.
- GIT.
- Kidneys.



Tight junctions

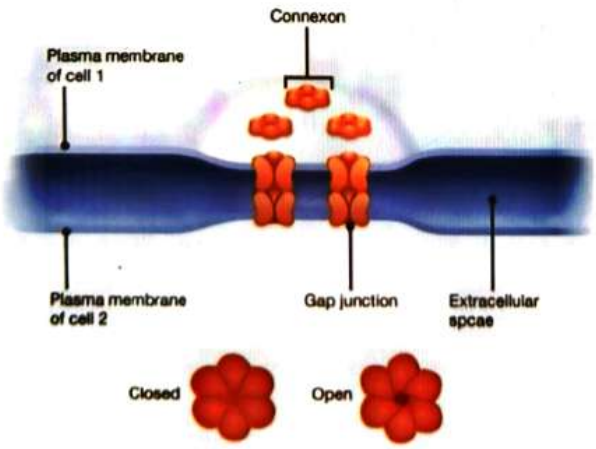
Clinical significance :

- Claudin -16 mutation (In Kidneys) : Familial hypomagnesemia hypercalciuria (↑ Ca^{2+} excretion → Renal stones).
- Regulates paracellular movements (Sideways/lateral movement).



Paracellular movement of tight junction

GAP JUNCTION :



Gap junction

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Location :

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Cells requiring rapid communication : Gap of 3-4 nm.

- Heart (Cardiac muscle) : Abundant
- Nervous (Nerve fibres).

Protein :

- Integral protein : Connexon.
- Subunits : Connexin.

Clinical significance :

- Charcot marie Tooth disease :
  - Mutation of connexin-32.
  - Affects neurons & nerve fibres (Peripheral sensory neuropathy).
- Idiopathic atrial fibrillation :
  - Mutation of connexin-40.
  - Affects heart.

CELL TO BASAL LAMINA JUNCTION :

Types :

- Hemidesmosomes : Attached to intermediate filaments.
- Focal adheshions : Attached to actin.

Clinical significance :

Metastatic : Cancer cells disrupt cell to basal lamina junction.

Feedback